METHOD FOR PERFORMANCE BENCHMARKING FOR PARALLEL ARCHITECTURES

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Title: METHOD FOR PERFORMANCE BENCHMARKING FOR PARALLEL ARCHITECTURES

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Abstract

A method for benchmarking the performance of computer systems with different bus architectures and clock speeds by calculating a normalized performance metric. The method comprises: defining a reference architecture with a reference bus width and reference clock speed; for a test system with a test bus width and test clock speed, calculating a "Parallelism-Adjusted Clock Speed Equivalent" (PACSE) metric by multiplying the test clock speed by the ratio of the test bus width to the reference bus width. This method provides a standardized way to compare the raw data throughput capabilities of a low-clock-speed, massively parallel system with a high-clock-speed, serial or narrow-parallel system, enabling fair and meaningful performance comparisons.

Background of the Invention

For decades, processor performance has been compared using the clock speed (in MHz or GHz) as a primary metric. This was a reasonable proxy when most systems shared a similar, narrow bus architecture (e.g., 32-bit or 64-bit). However, with the advent of massively parallel bus architectures where the bus width can be 256-bit, 1024-bit, or wider, a direct comparison of clock speeds is no longer meaningful. A 1 GHz processor with a 1024-bit bus has far greater data throughput than a 4 GHz processor with a 64-bit bus, but this is not evident from clock speed alone. A new benchmarking method is needed to account for the impact of parallelism on performance.

Summary of the Invention

The present invention is a method for performance benchmarking that normalizes the data throughput of different computer architectures into a single, comparable metric. The method allows for a fair comparison between systems that achieve performance through high clock frequency and systems that achieve performance through massive parallelism.

The method begins by establishing a baseline or reference architecture, typically a standard 64-bit bus. To benchmark a new system, its two key parameters are identified: its bus width and its clock speed. The core of the invention is the calculation of a normalized metric, such as a "Parallelism-Adjusted Clock Speed Equivalent" (PACSE). This is calculated with the formula: 'PACSE = (Test System Bus Width / Reference Bus Width) * Test System Clock Speed'.

For example, using a 64-bit reference, a system with a 256-bit bus running at 1 GHz would have a PACSE of '(256/64) * 1 GHz = 4 GHz'. This indicates it has the equivalent data throughput of a 4 GHz system that uses the 64-bit reference architecture. This method provides a clear, quantitative way to demonstrate the performance advantages of parallelism, even at lower clock speeds.

Statement of License Grant to the United States Government

The inventor, Reinaldo Javier Aguiar Marcano, hereby grants to the United States Federal Government and its military and defense agencies a perpetual, royalty-free, non-exclusive

license to practice and use the invention described herein for any and all governmental, defense, and non-commercial purposes. This license explicitly excludes any and all state and local governments.

Brief Description of the Drawings

FIG. 1 is a flowchart illustrating the steps of the performance benchmarking method.

Detailed Description of the Invention

Referring to **FIG. 1**, the steps of the benchmarking method are shown. The process begins with defining a reference architecture, including a reference bus width and clock speed (101). Next, the parameters for the test system are identified, including its test bus width and clock speed (102). The core calculation is then performed (103), where the Parallelism-Adjusted Clock Speed Equivalent (PACSE) is computed by multiplying the test clock speed by the ratio of the test bus width to the reference bus width. The resulting normalized metric (104) is then used to compare the test system to the reference system or other test systems (105), providing a fair and meaningful benchmark of their data throughput capabilities.

Claims

What is claimed is:

1. A method for benchmarking the performance of a computer system, the method comprising the steps of:
    - identifying a test bus width and a test clock speed of a test computer system;
    - defining a reference bus width for a reference computer system; and
    - using a computer to calculate a normalized performance metric by multiplying the test clock speed by the ratio of the test bus width to the reference bus width.
2. The method of claim 1, wherein the normalized performance metric is a Parallelism-Adjusted Clock Speed Equivalent.
3. The method of claim 1, wherein the reference bus width is 64 bits.
4. The method of claim 1, wherein the test bus width is greater than 128 bits.

5. The method of claim 1, further comprising the step of comparing the normalized performance metric of the test computer system with a clock speed of the reference computer system to determine relative performance.

6. The method of claim 1, wherein the test bus is a massively parallel optical bus.

Drawings

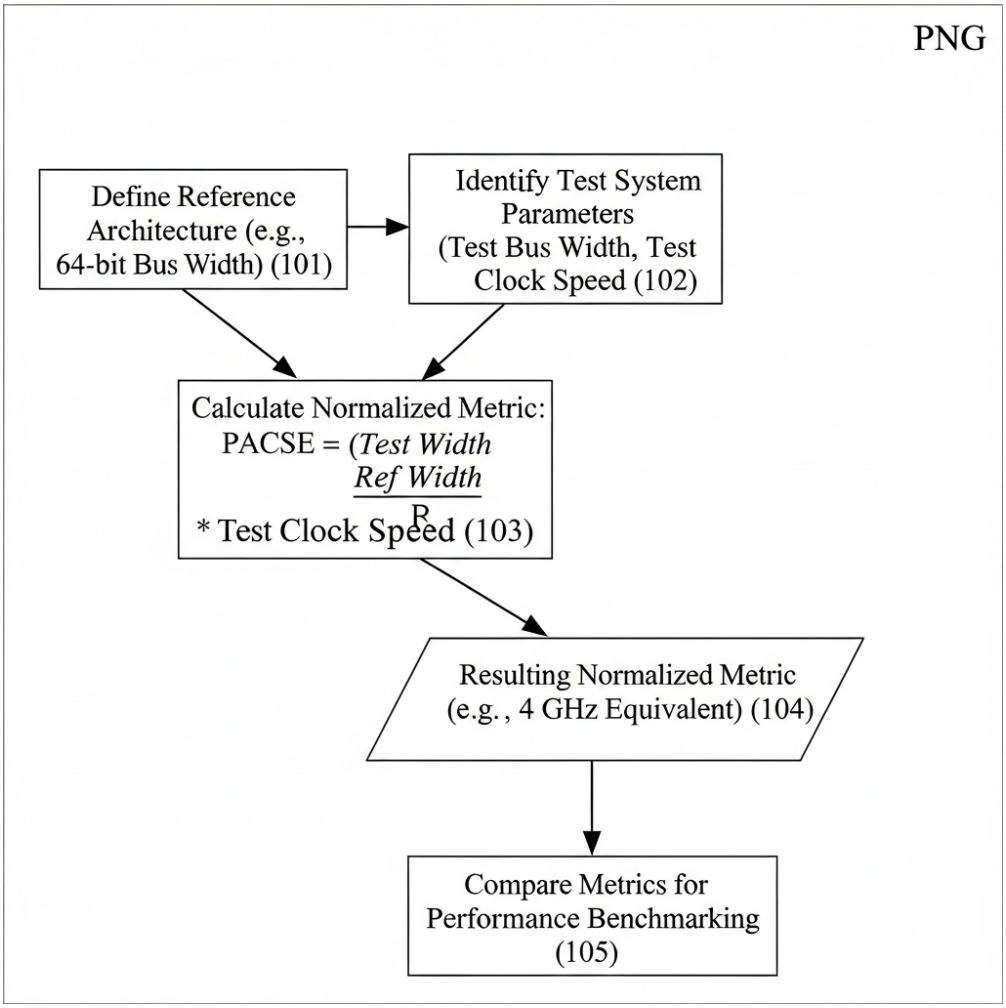


Figure 1: Flowchart illustrating the steps of the performance benchmarking method.